Supplementary Materials for "What Standard Evaluation Misses in Small-Model RLVR: Evidence from 30 Countdown Training Runs"

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Tables S1–S3 report greedy benchmark accuracy for the base model and every one of the 30 step-300 checkpoints, for every seed, method, benchmark, and prompt; no seed or benchmark was omitted. Table S4 reports sampled pass@k for every checkpoint. A dagger marks early-short runs (mean response length over steps 90–100 at most 100 tokens). Prompts, scorers, and sampling settings are described in the article; Supplementary Notes S1–S4 follow the tables.

Table S1: Flexible accuracy (%) for the base model and all 30 step-300 checkpoints. Std. denotes the benchmark-specific standard prompt; Think denotes the Countdown-style think prompt. †Early-short run.

	GSM8K		MATH		BBH		MMLU-Pro	
Model	Std.	Think	Std.	Think	Std.	Think	Std.	Think
Base	72.8	51.6	55.3	49.9	35.3	36.5	32.8	27.2
PPO s7	70.4	70.7	53.3	45.2	34.9	40.5	32.5	27.3
PPO s8†	48.3	0.0	55.0	0.0	32.0	0.0	29.5	0.0
PPO s9	74.1	71.2	54.0	48.1	33.6	37.8	32.7	25.4
PPO s10†	44.0	0.0	8.8	0.0	20.1	0.0	9.7	0.0
PPO s11	74.0	68.4	54.4	43.0	32.9	38.1	31.7	25.7
PPO s12	71.9	71.9	52.4	49.1	34.2	36.9	32.7	25.3
PPO s13	74.6	72.2	53.7	50.2	31.8	39.3	33.9	30.1
PPO s14	74.8	21.5	54.9	10.3	31.3	7.1	32.3	6.1
PPO s15	73.2	65.6	54.6	48.9	31.4	37.2	32.6	24.5
PPO s16†	69.0	0.2	52.5	0.9	23.1	0.4	31.1	0.2
GRPO s7	72.3	68.2	54.2	49.7	31.9	39.1	32.7	25.1
GRPO s8†	55.5	7.4	38.7	14.8	33.0	35.2	29.6	24.9
GRPO s9†	71.5	34.0	53.3	35.4	35.8	31.4	33.4	19.6
GRPO s10†	63.2	4.2	52.0	13.6	33.4	35.1	32.2	18.3
GRPO s11	73.0	64.6	55.3	51.1	32.9	38.1	32.3	22.0
GRPO s12†	71.3	7.1	53.1	19.2	34.3	34.5	32.2	22.0
GRPO s13	75.0	72.8	54.5	41.2	24.9	39.3	33.5	28.9
GRPO s14†	74.0	70.7	54.1	49.8	33.4	41.6	32.1	25.6
GRPO s15†	73.7	10.5	53.6	16.4	35.7	34.8	32.9	24.9
GRPO s16†	66.0	5.7	51.2	11.3	33.9	33.9	33.5	18.6
DAPO-style s7	73.2	71.3	52.5	48.7	36.0	42.3	32.0	26.7
DAPO-style s8†	68.1	8.7	52.0	15.9	34.7	35.9	31.1	20.3
DAPO-style s9†	73.5	6.2	52.9	11.7	36.9	34.4	32.3	15.1
DAPO-style s10†	70.1	13.0	46.9	17.0	32.5	36.4	30.6	25.4
DAPO-style s11	73.2	65.8	54.5	39.7	29.6	35.5	30.0	19.7
DAPO-style s12†	69.7	11.3	53.1	18.2	34.7	36.0	32.9	23.6
DAPO-style s13†	71.7	12.9	52.5	18.8	35.7	28.7	33.4	16.1
DAPO-style s14†	74.8	7.9	54.0	12.8	33.6	34.8	33.4	12.1
DAPO-style s15†	73.5	66.8	53.9	47.2	33.2	39.0	31.3	24.2
DAPO-style s16†	58.7	23.9	45.2	20.5	32.8	34.7	29.9	24.8

Table S2: Strict accuracy (%) for the base model and all 30 step-300 checkpoints. †Early-short run.

	GSM8K		MATH		BBH		MMLU-Pro	
Model	Std.	Think	Std.	Think	Std.	Think	Std.	Think
Base	33.3	48.3	55.3	47.8	35.0	35.1	32.8	27.2
PPO s7	59.7	70.7	53.2	43.8	34.8	40.5	32.5	27.3
PPO s8†	36.2	0.0	54.9	0.0	31.9	0.0	29.4	0.0
PPO s9	68.2	71.2	54.0	46.4	33.5	37.8	32.6	25.4
PPO s10†	42.6	0.0	8.4	0.0	20.1	0.0	9.7	0.0
PPO s11	73.2	68.3	54.4	42.6	32.9	38.1	31.7	25.7
PPO s12	68.6	71.9	52.4	48.3	34.1	36.9	32.7	25.3
PPO s13	71.1	72.2	53.7	48.8	31.3	39.2	33.9	30.1
PPO s14	73.7	21.3	54.8	10.1	31.1	7.1	32.3	6.1
PPO s15	72.4	65.6	54.5	48.1	31.2	37.2	32.6	24.5
PPO s16†	67.6	0.2	52.5	0.8	22.9	0.4	31.1	0.2
GRPO s7	69.7	67.3	54.1	49.0	31.9	39.1	32.7	25.1
GRPO s8†	27.7	7.4	37.0	14.6	32.3	35.1	29.6	24.9
GRPO s9†	54.4	33.8	53.2	33.7	35.5	31.3	33.3	19.5
GRPO s10†	43.2	4.2	51.3	12.7	32.6	35.1	32.2	18.3
GRPO s11	71.2	60.9	55.3	50.1	32.6	38.1	32.3	22.0
GRPO s12†	50.0	7.1	53.1	18.9	33.9	34.5	32.2	22.0
GRPO s13	74.5	72.5	54.5	40.6	24.8	39.3	33.5	28.9
GRPO s14†	62.7	70.7	54.0	48.5	33.1	41.6	32.1	25.5
GRPO s15†	51.5	10.5	53.6	16.0	35.3	34.8	32.9	24.9
GRPO s16†	32.9	5.6	50.4	11.1	33.6	33.9	33.5	18.6
DAPO-style s7	69.6	71.3	52.5	48.5	35.8	42.3	32.0	26.7
DAPO-style s8†	60.3	8.7	51.8	14.4	33.0	35.9	31.0	20.3
DAPO-style s9†	58.9	6.2	52.8	11.1	36.7	34.4	32.3	15.1
DAPO-style s10†	51.8	13.0	46.8	16.8	32.4	36.4	30.6	25.4
DAPO-style s11	70.3	65.8	54.5	39.1	29.6	35.5	30.0	19.7
DAPO-style s12†	36.0	11.3	52.9	17.7	34.6	36.0	32.9	23.6
DAPO-style s13†	55.3	12.8	52.5	16.1	35.5	28.7	33.4	16.0
DAPO-style s14†	71.4	7.8	54.0	11.5	33.1	34.8	33.4	12.1
DAPO-style s15†	73.2	66.7	53.9	46.2	33.2	39.0	31.3	24.2
DAPO-style s16†	35.0	17.5	42.9	19.1	29.6	34.7	29.9	24.8

Table S3: Model- and checkpoint-level GSM8K flexible accuracy (%) for the post hoc prompt-component controls. Standard is the conventional GSM8K prompt; Original denotes the Countdown-style think prompt. †Early-short run.

Model/checkpoint	Standard	Original	No example	Example 17	No opening tag	No assistant prefill	Clean tags
Base	72.8	51.6	58.5	51.9	66.5	70.9	68.3
PPO s7	70.4	70.7	67.9	70.4	69.1	71.5	68.2
PPO s8†	48.3	0.0	0.0	0.0	0.0	0.0	14.1
PPO s9	74.1	71.2	71.7	71.1	72.7	74.5	71.7
PPO s10†	44.0	0.0	0.0	0.0	0.0	2.3	26.7
PPO s11	74.0	68.4	70.0	68.1	72.9	73.2	72.3
PPO s12	71.9	71.9	71.0	70.7	67.4	68.2	67.9
PPO s13	74.6	72.2	72.3	72.6	71.4	71.3	70.3
PPO s14	74.8	21.5	22.5	20.3	52.9	70.7	68.8
PPO s15	73.2	65.6	67.5	65.0	69.4	73.7	68.2
PPO s16†	69.0	0.2	0.2	0.2	0.0	1.2	6.4
GRPO s7	72.3	68.2	70.4	68.4	67.5	71.5	68.8
GRPO s8†	55.5	7.4	12.5	7.3	46.6	46.6	42.4
GRPO s9†	71.5	34.0	49.8	35.0	48.4	45.0	43.4
GRPO s10†	63.2	4.2	5.2	4.6	5.5	9.0	8.0
GRPO s11	73.0	64.6	70.2	63.8	69.1	67.7	61.1
GRPO s12†	71.3	7.1	9.5	8.3	53.5	58.3	53.2
GRPO s13	75.0	72.8	72.9	71.7	71.2	72.4	70.5
GRPO s14†	74.0	70.7	69.6	69.9	60.5	59.7	66.4
GRPO s15†	73.7	10.5	8.9	9.2	35.1	34.3	38.7
GRPO s16†	66.0	5.7	14.3	6.5	48.1	53.5	52.3
DAPO-style s7	73.2	71.3	71.5	71.8	70.8	72.5	70.7
DAPO-style s8†	68.1	8.7	7.4	8.6	18.8	19.5	16.7
DAPO-style s9†	73.5	6.2	20.5	6.6	16.9	13.1	21.6
DAPO-style s10†	70.1	13.0	15.5	12.9	27.7	32.7	25.1
DAPO-style s11	73.2	65.8	66.4	64.5	67.0	68.7	61.7
DAPO-style s12†	69.7	11.3	44.9	11.7	23.4	31.6	30.8
DAPO-style s13†	71.7	12.9	13.6	13.6	18.5	14.9	22.0
DAPO-style s14†	74.8	7.9	41.6	8.3	8.3	9.3	13.7
DAPO-style s15†	73.5	66.8	69.6	65.8	73.3	66.9	60.0
DAPO-style s16†	58.7	23.9	33.1	23.8	31.8	34.9	21.5

Table S4: Sampled pass@k (%) under standard prompts for the base model and all 30 step-300 checkpoints. Cap: share of samples that reached the 4,096-token limit. Base: the reference sampling run; four further MATH-500 sampling runs of the base model with other random seeds follow it. Lost/gained: MATH-500 items solved at least once in the reference run but never by the model, and the reverse. †Early-short run.

	MATH-500 ($n = 64$)				GSM8K ($n = 32$)			
Model	Flex. @1	Flex. @64	Cap	Lost/gained	Flex. @1	Flex. @32	Strict @1	Cap
Base	53.9	92.2	2.3	461 solved	70.6	95.9	37.1	0.2
Base, sampling run 2	53.9	90.8	2.3	12/5				
Base, sampling run 3	54.0	91.8	2.4	10/8				
Base, sampling run 4	53.9	92.2	2.3	7/7				
Base, sampling run 5	54.0	91.2	2.4	8/3				
PPO s7	52.9	89.2	4.5	23/8	69.9	95.8	58.4	0.5
PPO s8†	52.0	91.0	1.2	16/10	49.4	95.2	38.9	1.1
PPO s9	53.9	89.6	5.3	18/5	73.5	96.2	67.9	0.8
PPO s10†	8.0	73.6	78.5	95/2	19.5	88.0	18.2	71.7
PPO s11	54.2	89.4	5.9	22/8	73.4	96.4	72.1	0.9
PPO s12	53.4	90.2	5.1	17/7	70.8	95.9	66.9	0.7
PPO s13	53.9	91.6	5.7	14/11	73.4	96.5	68.4	0.6
PPO s14	54.6	89.8	3.2	17/5	73.6	96.5	71.8	0.5
PPO s15	54.3	91.2	5.2	12/7	73.1	96.1	71.9	0.8
PPO s16†	47.1	89.6	4.6	22/9	64.5	96.6	63.0	2.2
GRPO s7	52.7	89.4	5.1	20/6	72.1	96.6	68.6	0.6
GRPO s8†	37.8	88.0	2.7	26/5	50.9	95.1	26.6	0.6
GRPO s9†	53.3	90.2	2.4	18/8	69.8	96.4	54.5	0.2
GRPO s10†	49.7	89.8	2.0	22/10	61.6	96.6	44.2	0.5
GRPO s11	54.4	89.0	3.0	22/6	72.9	96.6	69.8	0.6
GRPO s12†	52.3	90.8	2.7	14/7	69.6	96.8	53.7	0.3
GRPO s13	54.0	90.4	7.6	16/7	72.4	96.6	71.9	2.9
GRPO s14†	53.9	90.2	2.3	17/7	72.0	96.5	62.2	0.4
GRPO s15†	52.4	88.6	3.1	24/6	71.2	96.5	45.8	0.3
GRPO s16†	49.2	90.8	2.2	15/8	63.8	96.8	31.5	0.2
DAPO-style s7	51.6	88.8	6.1	23/6	71.4	96.2	67.4	0.7
DAPO-style s8†	48.7	91.0	1.5	14/8	65.8	96.7	56.1	0.2
DAPO-style s9†	49.9	90.4	2.1	19/10	70.0	96.4	55.2	0.3
DAPO-style s10†	42.7	89.2	4.0	22/7	68.2	97.0	51.5	0.3
DAPO-style s11	52.0	90.0	6.5	20/9	71.6	96.2	67.2	3.0
DAPO-style s12†	50.3	91.0	2.2	15/9	69.4	97.2	42.3	0.2
DAPO-style s13†	52.9	87.4	1.8	28/4	70.2	96.9	55.9	0.2
DAPO-style s14†	52.6	90.8	2.3	13/6	71.4	97.0	66.0	0.3
DAPO-style s15†	54.5	89.8	3.7	19/7	73.1	97.0	72.8	0.5
DAPO-style s16†	42.3	88.2	1.5	28/8	57.2	92.8	37.2	0.1

Supplementary Note S1. Evaluation Environment and Data Identifiers

Chat wrapping and stopping. Each benchmark prompt below was passed as the content of one user message to the Qwen tokenizer's `apply_chat_template` method with `add_generation_prompt=True`. Greedy evaluation used vLLM with bfloat16 weights, seed 0, temperature 0, an 8,192-token generation limit, a 10,240-token context, and the model's default EOS stopping behavior. Pass@k used the same wrapping with seed 0, temperature 0.6, top-$p = 0.95$, and a 4,096-token cap. All training runs and the base model used Qwen model snapshot `989aa7980e4cf806f80c7fef2b1adb7bc71aa306`.

Hardware and software. The base model and all 30 step-300 checkpoints were evaluated on NVIDIA H100 NVL GPUs (driver 575.57.08) with vLLM 0.10.0, PyTorch 2.7.1+cu128, and Transformers 4.53.2, using one packaged set of frozen prompt files and one model per GPU. Generation ran on the cluster; scoring ran on the original workstation with the unchanged scorers. Every returned output file is checked against the SHA-256 recorded when it was generated. Countdown prompts were generated for the base model and all checkpoints on one two-RTX-4090 workstation with the same vLLM, PyTorch, and Transformers versions. The base model and all 30 checkpoints were sampled on one workstation with four RTX A6000 GPUs and vLLM 0.10.0; outputs were scored on the original machine with the unchanged scorer. Re-sampling the base model on the same hardware keeps the hardware and software common within each model–base contrast. Sampled pass@1 is an average over stochastic outputs, not the greedy accuracy reported in the main benchmark tables.

Checkpoint identifiers. The manifest `eval/CHECKPOINT_MANIFEST.json` identifies the 30 step-300 checkpoints by per-file and aggregate SHA-256 hashes; the evaluation package verifies every file against these hashes before generation.

Dataset identifiers. GSM8K used the 1,319-item test parquet distributed with the local Verl data preparation. Its upstream revision was not retained, but the exact evaluated file is vendored as `eval/data/gsm8k_test.parquet` with SHA-256 `f284e0ad015991d0059d265051a3c0915c5c1def15a8c01ac9a9142eeb87a180`. Exact comparison of every saved input/reference row with locally cached immutable snapshots recovered the other revisions: MATH used all seven test configurations from `EleutherAI/hendrycks_math` at `21a5633873b6a120296cce3e2df9d5550074f4a3` (5,000/5,000 rows); BBH used all 27 test configurations from `SaylorTwift/bbh` at `b5306be6f827cfafbb545ff5a51f96916029b0fd` (6,511/6,511 rows); and MMLU-Pro used the test split of `TIGER-Lab/MMLU-Pro` at `dd36ce4b34827164989f100331f82c5a29741747` (12,032/12,032 rows). The MATH-500 pass@k subset is an exact 500-item subset of that pinned MATH snapshot and is preserved as `eval/data/math500.jsonl`, SHA-256 `200806fb17234213e909649c09a14ce01ef836b32d`. The machine-readable manifest `eval/DATASET_MANIFEST.json` records canonical input digests, and `eval/verify_dataset_manifest.py` checks the archived files offline and can reconstruct the three revision-pinned datasets.

Archived code and outputs. The prompt builders are archived in `eval/eval_gsm8k_prompt_ablation.py` and `eval/eval_gsm8k_prompt_open_tag_ablation.py`. The variants were generated with the same package, hardware, decoding, and 8,192-token limit as the primary prompts, for the base model and all 30 checkpoints. The saved per-item files contain the raw response, extracted strict and flexible predictions, correctness flags, and finish reason so that extraction can be re-audited without regenerating model responses.

Supplementary Note S2. Additional Prompt Diagnostics

For the base model, removing the opening reasoning tag recovered 14.9 percentage points relative to the original wrapper; removing the complete prefill recovered 19.3 points. The latter condition differed from the standard prompt by 1.9 points; this difference cannot be attributed to the remaining prompt differences, because the paired exact McNemar p-value is 0.14, although this is not evidence that the two prompts are equivalent.

Early-short checkpoints failed under the think prompt in two different ways. The 15 early-short GRPO and DAPO-style checkpoints closed the reasoning block almost immediately (median response 13–250 tokens; at most 1.7% of items reached the 8,192-token limit). The three early-short PPO checkpoints instead produced unbounded output: 96.5–100% of items reached the limit, so flexible scoring credited almost none. PPO seed 10 also reached the limit on 34.1% of standard-prompt items, by repeating one word or a short phrase until the limit, which accounts for its low standard-prompt accuracy. Prompt changes interacted with these modes: the clean tag instruction reduced limit-reaching to 8.0% for PPO seed 10 and 61.3% for PPO seed 8, and removing the complete prefill reduced it from 68.2% to 1.3% for PPO seed 14, a run just above the early-short threshold.

Countdown without the training wrapper. We also posed the 1,024 Countdown test items as a plain request, without role lines, prefill, or tags, asking for the final equation on a last line `Answer: <equation>` (prompt in Appendix A of the article), with greedy decoding. Checkpoints that did not shorten early averaged 49.9% under the training wrapper and 44.4% under the plain prompt (9 of 12 significantly lower, 2 higher; exact McNemar tests), and early-short checkpoints averaged 26.2% and 22.6% (9 of 18 lower, 1 higher). Under the plain prompt, checkpoints that did not shorten early gave a final answer on a median 63.8% of items, versus 98.2% under the wrapper. The two largest gains came from explosion checkpoints: PPO seeds 14 and 8 scored 0.0% under the wrapper, where they produced 8,192 tokens of degenerate text, but 44.1% and 21.1% under the plain prompt, so their degeneration was specific to the training wrapper. GRPO seed 8, by contrast, answered every plain-prompt item with "Okay." The base model scored 11.2% under the wrapper and 1.8% under the plain prompt.

Supplementary Note S3. Additional Training Diagnostics

Clip fractions around shortening. In a post hoc analysis aligned on the step at which each early-short run first fell below 100 tokens, the median lower clip fraction rose from 0.07% of response tokens 11–15 steps earlier to 0.49% in the preceding five steps and 0.91% in the following five, while the upper clip fraction stayed at 0.03–0.08%. In steps 1–5, early-short and other runs did not differ (median 0.015% each), so this diagnostic accompanied shortening rather than anticipating it. Early-short runs then ended in either entropy regime, from 0.06 nats to more than 5 nats at step 300.

KL loss in resumed runs. The seven explosion runs were resumed from a checkpoint saved before entropy onset (steps 50–250) and trained to step 300 without KL, with $\beta = 0.001$, and with $\beta = 0.05$ (the 0.001 arm covers five runs), one run per arm. Onset is the first step after step 20 from which entropy stays above 1 nat for five steps. The resumed no-KL copies did not repeat their originals exactly: onset moved by -35 to $+24$ steps, and PPO seed 14 stayed below 1 nat. With $\beta = 0.001$, entropy exploded in all five runs (PPO seed 14 only with KL), with onset -4 to $+31$ steps from the no-KL copies; the penalty was at most about 0.003, against a mean absolute policy loss of 0.007–0.12. With $\beta = 0.05$, onset came 34–105 steps later in five runs, 27 steps earlier in DAPO-style seed 15, and only with KL in PPO seed 14; step-300 entropy was 0.63–3.20 nats, versus 5.14–9.44 for the six no-KL copies whose entropy exploded, and six of seven runs still ended above 1 nat. Responses also shortened, and step-300 Countdown accuracy was 2.8–29.7 points lower in the three group-relative runs.

Supplementary Note S4. Additional Coverage Details

Supplementary Table S4 reports sampled pass@k for every checkpoint. The base model solved 461 of the 500 MATH-500 items at least once in 64 samples. Apart from PPO seed 10, each checkpoint failed on 12–28 of these items in all 64 samples and solved 4–11 items the base model never solved, so the coverage loss is a net change rather than a uniform shift. Item-level counts are noisy, however: resampling the base model with four other seeds lost 7–12 and gained 3–8 of the same items. Sampling used a 4,096-token cap. The base model reached the cap on 2.3% of MATH-500 samples, and the checkpoints that did not

shorten early did so on 3.0–7.6%; the early-short checkpoints reached it less often (1.2–4.6%) yet lost similar coverage, so truncation alone does not account for the deficit. PPO seed 10 reached the cap on 78.5% of MATH-500 and 71.7% of GSM8K samples, consistent with the repetition seen under greedy decoding.